

Neurons and neuroglia

The written notes for the neurons and neuroglia walkthrough. Week 4 competencies 7 to 11.

To print: these notes as a two-column PDF. Already printed the Week 4 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

Organization of the nervous system

- The central nervous system (CNS) is the brain and spinal cord. It is the integrating center: it receives incoming information, combines it with everything else going on, and selects a response.
- The peripheral nervous system (PNS) is all the neural tissue outside the CNS: the cranial nerves, the spinal nerves, and their ganglia. It carries signals between the CNS and the rest of the body, in both directions.
- The sensory division, also called the afferent division, carries information toward the CNS from receptors in the skin, muscles, eyes, ears and internal organs. Most signals from the organs never reach awareness.
- The efferent division carries signals away from the CNS to effectors, which are muscles and glands.

Efferent branch	What it controls	Example
Somatic motor division	Skeletal muscle	Picking up a pencil (voluntary); many reflexes
Autonomic division (visceral nervous system)	Smooth muscle, cardiac muscle, glands, some adipose tissue	Heart rate speeding up (involuntary)

- The autonomic division has two branches that often have opposite effects on the same organ. The organ's activity at any moment reflects the balance between them.
- Sympathetic branch: fight-or-flight. Prepares the body for stress and effort, for example by raising heart rate and sending more blood to working muscles.
- Parasympathetic branch: rest-and-digest. Supports maintenance functions such as digestion, and slows the heart.
- Enteric nervous system: a network of neurons in the wall of the digestive tract that controls how the gut moves and what it secretes. It can detect a stimulus, integrate it and send out a response on its own, and the autonomic division normally adjusts it.

Afferent

Carrying toward the CNS. The sensory division.

Efferent

Carrying away from the CNS, toward effectors.

Effector

A muscle or gland that carries out the response.

Integrating center

The part of a pathway that receives incoming signals, sums them and selects a response. In the nervous system, the CNS.

Functional classes and shapes of neurons

By direction	What it does	Where
Sensory (afferent) neuron	Carries information toward the CNS	Receptor to CNS
Interneuron	Connects other neurons; carries out integration. Roughly 99 percent of neurons are interneurons.	Entirely inside the CNS
Efferent (motor) neuron	Carries signals away from the CNS to effectors	CNS to effector

- A typical neuron has dendrites, short branches that receive signals, and one axon, a long fiber that carries signals away from the cell body.
- Neurons are sorted by shape by counting the processes that leave the cell body.

Shape	Processes	Where it is found
Multipolar	Many dendrites and one axon	Most interneurons of the brain and spinal cord, and efferent neurons such as somatic motor neurons (cell bodies in the ventral horn)
Bipolar	One process on each side of the cell body	Retina and olfactory epithelium, where they carry signals in the pathways for vision and smell
Pseudounipolar	One stalk that splits into a peripheral branch and a central branch	Sensory neurons for touch, pain and stretch; cell bodies in the dorsal root ganglia

- In a pseudounipolar neuron the action potential travels along one continuous axon, from the peripheral branch straight into the central branch, so the cell body is not in the conduction pathway. The cell body sits to one side in a dorsal root ganglion and still makes the proteins the neuron needs.
- Sensory fibers enter the spinal cord through the dorsal root and efferent fibers leave through the ventral root. If one dorsal root is cut, sensation in the area it serves is lost but voluntary movement is kept. Reflexes in that area are lost too, because the afferent half of each reflex pathway has been cut.

Functional regions of a neuron

1. Dendrites receive signals from other cells. One neuron may receive synaptic input from thousands of others. The signals here are graded potentials: small, local, variable in size, and they get smaller as they spread.
 2. The cell body (soma) holds the nucleus and makes nearly all of the proteins the neuron needs. Signals from every dendrite spread into it and add together, which is called summation.
 3. The trigger zone, the axon hillock and the initial segment of the axon, is where the summed graded potentials are integrated: if the sum reaches threshold, an action potential starts here. It has the highest density of voltage-gated sodium channels in the neuron, so it has the lowest threshold and reaches threshold before any other region. In many sensory neurons the trigger zone lies near the receptor endings instead.
 4. The axon carries action potentials. An action potential is all or none: voltage-gated channels regenerate it at full size as it travels, so it arrives at the terminals the same size as it started. A motor axon from the lower spinal cord to a muscle of the foot can be close to a meter long.
 5. The axon terminals release neurotransmitter. When an action potential depolarizes the terminal, voltage-gated calcium channels open, calcium moves into the terminal down its concentration gradient, and the rise in calcium causes vesicles to fuse with the membrane and release neurotransmitter by exocytosis.
- A stronger stimulus does not make a bigger action potential. It makes each sensory neuron fire more often, and it activates more neurons.
 - An input on the cell body next to the trigger zone has a bigger effect there than an identical input at the far tip of a dendrite, because it has less distance over which to get smaller.
 - If the calcium channels at the terminal are blocked, the action potential arrives normally but no calcium enters, so no neurotransmitter is released and the signal stops at the terminal.

Graded potential

A small, local change in membrane potential whose size varies with the strength of the input and which gets smaller with distance.

Action potential

A large, all-or-none signal that travels the length of the axon without getting smaller.

Trigger zone

The axon hillock and initial segment, where the summed graded potentials either reach threshold and start an action potential or do not.

Synapse

The junction between an axon terminal and the next cell: a neuron, a muscle fiber or a gland cell. The gap between them is the synaptic cleft.

Glial cells

Glial cell	Where	Functions
Astrocyte	CNS	Most common CNS glial cell. Regulates the extracellular fluid around neurons: takes up excess potassium after neurons fire, removes neurotransmitter such as glutamate from synapses, and helps deliver nutrients from the blood to neurons. Its end feet release signals that induce and maintain the tight junctions of the blood-brain barrier.
Oligodendrocyte	CNS	Forms myelin. One oligodendrocyte wraps segments of several axons.
Microglia	CNS	Resident immune cells. They survey the surrounding tissue, and when they detect damage, debris or pathogens they become activated and remove them by phagocytosis.
Ependymal cell	CNS	Lines the ventricles and central canal. Specialized ependymal cells produce cerebrospinal fluid; cilia help keep it circulating.
Schwann cell	PNS	Forms myelin. One Schwann cell wraps one segment of one axon. Helps cut peripheral axons regrow.
Satellite cell	PNS	Forms a capsule around neuron cell bodies in ganglia and regulates the fluid around them. Closely related to Schwann cells but does not form myelin.

- Blood-brain barrier: the endothelial cells of brain capillaries are joined by tight junctions. Small lipid-soluble molecules (oxygen, carbon dioxide, alcohol) cross easily; water-soluble molecules (glucose, amino acids) depend on transport proteins. The barrier keeps out many toxins, and also many drugs.
- If astrocytes cannot remove potassium, extracellular potassium rises, the resting membrane potential becomes less negative, and neurons rest closer to threshold. This is one way astrocyte dysfunction is thought to contribute to some seizure disorders.
- Hydrocephalus: if cerebrospinal fluid cannot drain, it accumulates, the ventricles upstream of the block enlarge, and the rising pressure compresses brain tissue. The fluid volume stays constant only when production equals drainage.
- Because one oligodendrocyte myelinates many axons, losing one oligodendrocyte removes myelin from many axons.

When an axon is cut

1. The part of the axon separated from the cell body degenerates, because it no longer receives proteins from the cell body.
2. In the PNS, Schwann cells survive, help clear debris, line up to form a tube and release growth factors, and the axon can regrow along that path, at roughly a millimeter a day.
3. In the CNS, oligodendrocytes and astrocytes release molecules that inhibit regrowth, and astrocytes form a glial scar.

Myelin and what happens when it is lost

- Myelin is the plasma membrane of a glial cell wound around the axon many times. Membrane is mostly lipid and ions cannot cross the lipid bilayer on their own, so myelin acts as electrical insulation.
- Myelin covers the axon in segments. The short uncovered gaps between them are the nodes of Ranvier.

- Voltage-gated sodium channels are densely clustered at the nodes, with very few in the membrane under the myelin.
- The slow step in conduction is the opening of voltage-gated channels that regenerates the action potential, not current flowing along the inside of the axon. Myelin reduces current leak, so current from one node reaches the next node still strong enough to bring it to threshold and regenerate the action potential there.
- The action potential is regenerated only at the nodes, so it effectively jumps from node to node. This is saltatory conduction. More nodes over the same distance would mean more regeneration steps and slower conduction.
- In electrical terms, myelin raises membrane resistance (less current leaks out) and lowers membrane capacitance (less charge is needed to change the voltage). A wider axon also conducts faster, because it has less resistance to current flowing along the inside.

Direction	Motor protein	Cargo
Anterograde: cell body to terminals	Kinesin	Synaptic vesicles, new membrane proteins such as channels, mitochondria
Retrograde: terminals to cell body	Dynein	Worn-out membrane and organelles for recycling, growth factors released by the target cell

Speed	Rate	Carries
Fast transport	Up to about 400 mm per day	Vesicles and mitochondria
Slow transport	About 0.2 to 2.5 mm per day	Cytoskeleton proteins and dissolved enzymes

- At 400 mm per day, fast transport covers a 1 meter axon in about 2.5 days.
- The slow rate of slow transport helps explain why a cut peripheral axon regrows only about a millimeter a day: the new axon is built largely from cytoskeleton proteins that arrive by slow transport.
- If outward transport is blocked at one point, outward cargo accumulates on the cell body side of the block. If inward transport is blocked, returning cargo accumulates on the terminal side. If dynein fails, worn-out material and growth factors no longer return to the cell body, and over time the neuron degenerates.
- Retrograde transport is also a route of infection. Rabies virus is taken up at motor neuron terminals in muscle and carried back along the axon. The cell bodies of these motor neurons are in the spinal cord, so the virus arrives directly in the CNS and then spreads from neuron to neuron to the brain. Tetanus toxin travels the same retrograde route.
- Herpes viruses use both directions: retrograde transport carries them to a sensory neuron cell body, where they remain dormant, and years later anterograde transport carries them back out along the same axon, which is why a cold sore returns in the same spot.

Axon	Conduction speed
Large, myelinated	Up to about 120 m/s
Thin, unmyelinated	About 0.5 to 2 m/s

Demyelination

1. The myelin over a stretch of axon is destroyed and the axon membrane underneath is exposed.
2. Current leaks out across the bare stretch.
3. Because that membrane was covered before, it has very few sodium channels to regenerate the action potential.
4. If the stretch is long enough, the current fades before the next node reaches threshold, and the action potential stops. This is conduction block.

Disease	Myelin attacked	Pattern
Multiple sclerosis	CNS myelin, formed by oligodendrocytes	Symptoms that seem unrelated (vision, sensation, strength, balance) depending on which axons lose myelin. They come and go as areas become inflamed and then partly recover. A rise in body temperature can briefly bring barely conducting axons into conduction block.
Guillain-Barré syndrome	PNS myelin, formed by Schwann cells	In its most common form, an immune attack on peripheral myelin that often follows an infection. Weakness spreads up the legs and reflexes are lost; conduction is slowed but the response size is near normal, so the axons are largely intact. Most people recover over weeks to months because Schwann cells remyelinate axons effectively.

Axonal transport

- Nearly all proteins are made in the cell body. Almost nothing is made in the axon or terminals, so enzymes, membrane and vesicles have to be transported there from the cell body.
- Cargo travels along microtubules that run the length of the axon. Motor proteins carry it step by step and hydrolyze one ATP for each step. If the microtubules are disassembled, transport stops in both directions.